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Ruei-Che Chang

I create novel interactive and enabling systems that aim to empower people with varying abilities and experiences to equivalently and independently make and consume physical and digital properties. My research interests span Human-AI Interaction, Mobile Interaction, Digital Fabrication, Mixed Reality and Accessibility.

Education

2020–present **National Taiwan University**, *Taipei, Taiwan*.
Visiting Student at Interactive Graphics Lab.
Advisor: Bing-Yu (Robin) Chen.
Collaborators: Anhong Guo(2021 Sep-Present), Shang-Tse Chen(2021 Jun-Nov), Tom Yeh(2021 Jan-May).

2019–2021 **Dartmouth College**, *New Hampshire, Hanover*.
M.S. in Computer Science (Research-based), with 75% Tuition supported by department.
Advisor: Xing-Dong Yang. Thesis Committee: Bing-Yu (Robin) Chen, Xia Zhou.
Thesis: Adaptive Add-on Circuit Components for People who are Blind or Have Low Vision.

2014–2018 **National Cheng Kung University**, *Tainan Taiwan*.
B.S. in Electrical Engineering.

Publications (All are full papers, *denote equal contribution)

In Conference Proceedings

2021 [C.6] **Ruei-Che Chang**, Chih-An Tsao, Fang-Ying Liao, Seraphina Yong, Tom Yeh, Bing-Yu Chen, "Daedalus in the Dark: Designing for Non-Visual Accessible Construction of Laser-Cut Architecture." *In Proceedings of the ACM Symposium on User Interface Software & Technology* (**UIST'21**).

[C.5] **Ruei-Che Chang***, Wen-Ping Wang*, Chi-Huan Chiang, Justin Luo, Te-Yen Wu, Zheer Xu, Bing-Yu Chen, Xing-Dong Yang, "AccessibleCircuits: Adaptive Add-On Circuit Components for People with Blindness or Low Vision." *In Proceedings of the ACM Conference on Human Factors in Computing Systems* (**CHI'21**).

2020 [C.4] **Ruei-Che Chang***, Chi-Huan Chiang*, Shuo-wen Hsu, Chih-Yun Yang, Da-Yuan Huang, Bing-Yu Chen, "TanGo: Exploring Expressive Tangible Interactions on Head-Mounted Displays." *In Proceedings of the ACM Conference on Spatial User Interface* (**SUI'20**).

[C.3] **Ruei-Che Chang***, Yi-Shyuan Chiang*, Yi-Lin Chuang, Shih-Ya Chou, Hao-Ping Lee, I-Ju Lin, Jian Hua Jiang Chen, Yung-Ju Chang, "Exploring the Design Space of User-System Communication for Smart home Routine Assistants." *In Proceedings of the ACM Conference on Human Factors in Computing Systems* (**CHI'20**).

[C.2] Kai-Chieh Huang, Chen-Kuo Sun, Da-Yuan Huang, Yu-Chun Chen, **Ruei-Che Chang**, Shuo-wen Hsu, Chih-Yun Yang, Bing-Yu Chen, "Glissade: Generating Balance Shifting Feedback to Facilitate Auxiliary Digital Pen Input." *In Proceedings of the ACM Conference on Human Factors in Computing Systems* (**CHI'20**) **[Honorable Mention Award, Top 5%]**.

2019 [C.1] Chi Wang, Da-Yuan Huang, Shuo-Wen Hsu, Chu-En Hou, Yeu-Luen Chiu, **Ruei-Che Chang**, Jo-Yu Lo, Bing-Yu Chen, "Masque: Exploring Lateral Skin Stretch Feedback on the Face with Head-Mounted Displays." *In Proceedings of the ACM Symposium on User Interface Software & Technology* (**UIST'19**).

Awards

2020 **Honorable Mention Award for Glissade [C.2]** at *ACM CHI 2020.*

2019 Taiwan internship scholarship from the Ministry of Education (acceptance rate 21.53%).

2017 2nd award at Taiwan Intelligent Robot Contest (30 participants).

Invited Talks/Hosts

Note: TAICHI is the top domestic HCI conference in Taiwan.

TAICHI'21 Invited to co-host a panel discussing research culture with 3 international PhD students.

TAICHI'21 Invited to present *Daedalus*[*C.6*] and *AccessibleCircuits*[*C.5*] at Best of HCI session.

Reviewer

CHI	'22
UIST	'21
LBW at CHI	'20 '21

Research Experience

Sep 2021 – present **Human-AI Lab, University of Michigan**
Remote research intern, advised by Anhong Guo.
- Developing an AI system to transform visual properties of existing 360 videos into immersive and interactive haptic-audio alternatives for blind or low-vision people. We target at UIST'22.

Jun 2021 – Nov 2021 **AI Security Lab, National Taiwan University**
Project Collaborator, advised by Shang-Tse Chen.
- Developing a website (including front and back end) to collect data of using our proposed audio CAPTCHAs from 120 online general users; Conducting an online usability test of our proposed CAPTCHAs with 12 blind users. In submission to IEEE Security & Privacy'22.

Jun 2020 – present **Interactive Graphics Lab, National Taiwan University**
Visiting student, advised by Bing-Yu (Robin) Chen and Tom Yeh.
- Leading weekly two-hours group discussion (15 master students) on different HCI topics; Mentoring master students on brainstorming and scoping research projects.
- Designing and conducting a study to identify accessibility challenges of laser-cut assembly; Implementing a Fusion 360 plug-in to generate tactile aids for input laser-cut model; Conducting an evaluation for tactile aids, and leading video and qualitative analysis [C.6].
- Designing and conducting a user study with blind and sighted groups to understand (non-)visual strategies of laser-cut assembly; Conducting inter-coder video analysis; In submission to CHI'22
- Collecting brain signal data using fNIRs on VR tasks, and analyzing data by MATLAB.

Feb 2019 – Aug 2019 ***Research Assistant, advised by Bing-Yu (Robin) Chen.***
Using Unity to implement six tablet applications for Glissade[C.2], and an authoring tool for designer to specify haptic feedback on a tablet for Masque [C.1].
- Integrating arduino software and electronics; Implementing two VR applications; Designing and conducting an user study to get preliminary feedback of using TanGo [C.4].

Sep 2019 – Sep 2020 **XDiscovery Lab, Dartmouth College**
Graduate student, advised by Xing-Dong Yang.
- Designing and fabricating conductive 3D-printed adaptions full of tactile aids for existing circuit components (via Fusion 360), which can interact with smartphone through a conductive breadboard extension casing; Designing and conducting a study to identify accessibility challenges of circuit prototyping, and an another study to evaluate designed adaptations [C.5].
- Collecting and labeling 500+ images of breadboard circuits as training data of Mask-RCNN. Implementing computer vision algorithms to segment breadboard holes and electronics pins, and visualize on-screen AR helps for students through an Android smartphone. (Manuscript Unpublished)

Sep 2018 – Apr 2019 **Mobile and Ubiquitous Interaction Lab, National Yang Ming Chiao Tung University**
Research Assistant, advised by Yung-Ju (Stanley) Chang.
- Designing and executing an experiment using methods: WoZ, speed dating, user enactment and contextual inquiry, to understand human-AI communication in smart home [C.3].

Student Mentoring

2021– present	**Zhao-Xian Ding**, *Master student*, National Taiwan University. (In preparation to UIST'22)
2021– present	**Fang-Ying Liao**, *Master student*, National Taiwan University. (In preparation to UIST'22)
2021 Jan–Apr	**Chih-An Tsao**, *Master student*, National Taiwan University. (Co-author [C.6], UIST'21)

Work Experience

2018 **Mandatory Military Service**, Taiwan.

2018 **Undergraduate research intern** at *Industrial Technology Research Institute* (*ITRI*), Hsinchu, Taiwan.

Skills

Programming	Java, C#, Python, JavaScript, ROS, OpenCV, MongoDB, Swift, LaTeX
Prototyping	Arduino, 3D-printing, Fusion 360, Laser-cutting
Platforms/IDE	Unity3D, Android Studio, Fusion 360, Xcode

References

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